

BARRAGE TEMISCAMINGUE, QC, Canada

WMO: 717320

Lat: **46.7097N**Lon: **79.1010W**Elev: **181**StdP: **99.16**Time Zone: **-5.00 (NAE)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-23.2	-29.5	0.3	-25.7	-26.5	0.3	-22.5	8.3	-12.0	7.6	-12.3	2.3	10	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.7	20.4	28.0	19.5	26.2	18.5	22.1	26.8	21.1	25.3	20.2	24.0	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.7	24.0	19.7	14.8	23.0	18.8	14.0	22.2	65.6	26.9	62.2	25.3	58.8	24.1	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.3	5.5	-29.7	32.3	2.9	1.6	-31.7	33.5	-33.4	34.4	-35.1	35.3	-37.2	36.4		
(5)			-29.8	24.2	2.9	1.5	-31.9	25.2	-33.6	26.1	-35.2	26.9	-37.2	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-10.7	-9.0	-3.6	3.9	11.2	16.4	19.6	18.8	14.7	7.7	0.8	-6.4
	DBStd	11.67	7.21	6.29	6.34	4.57	4.23	3.68	2.94	2.96	4.08	4.43	4.97	6.31
	HDD10.0	2717	642	532	422	189	36	2	0	0	7	102	278	508
	HDD18.3	4879	901	766	679	433	223	79	19	31	124	331	527	766
	CDD10.0	1020	0	0	1	6	74	194	297	272	146	30	1	0
	CDD18.3	140	0	0	0	0	3	21	58	44	13	1	0	0
	CDH23.3	1226	0	0	1	3	88	236	491	308	93	5	0	0
	CDH26.7	296	0	0	0	0	19	56	132	70	19	0	0	0
Wind	WSAvg	2.5	2.8	2.8	2.9	2.9	2.6	2.2	2.1	2.1	2.2	2.6	2.8	2.7
Precipitation	PrecAvg	921	63	51	60	70	80	81	92	83	92	94	84	69
	PrecMax	1103	108	107	216	130	140	152	189	148	157	172	149	143
	PrecMin	689	23	20	0	26	18	27	28	24	43	49	35	31
	PrecStd	108	22	23	41	25	33	32	44	34	32	37	27	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	7.0	15.0	23.0	29.1	30.4	32.3	31.2	29.3	23.5	15.0	8.4
		MCWB	5.4	4.1	8.0	13.8	18.5	20.5	21.9	21.1	21.2	16.2	11.0	6.7
	2%	DB	3.1	3.7	10.2	18.1	25.7	28.3	29.9	28.6	25.9	19.9	11.7	5.4
		MCWB	2.2	1.8	5.2	9.6	15.6	19.3	20.6	20.2	19.6	15.2	9.3	4.3
	5%	DB	1.5	1.9	7.1	14.7	23.1	26.2	28.0	26.6	23.6	16.9	9.5	3.4
		MCWB	0.7	0.5	3.7	7.9	14.1	18.2	19.6	19.1	18.1	13.2	7.7	2.4
	10%	DB	0.0	0.2	4.7	11.6	20.1	23.9	26.1	24.8	21.5	14.5	7.5	1.9
		MCWB	-0.9	-0.8	2.0	6.0	12.9	16.9	18.6	18.4	17.5	12.1	6.0	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	4.8	10.1	14.7	20.3	22.4	24.1	22.9	22.3	18.1	12.2	7.1
		MCDB	6.3	6.4	11.8	20.3	27.0	28.1	29.9	28.0	26.9	20.9	14.2	7.7
	2%	WB	2.4	2.1	6.2	11.1	17.6	21.1	22.3	21.6	20.8	15.6	9.7	4.4
		MCDB	3.0	3.4	9.1	16.4	22.6	26.1	26.7	26.0	23.8	19.0	11.2	5.1
	5%	WB	0.8	0.6	4.0	8.4	15.5	19.6	21.2	20.7	19.5	14.0	7.8	2.5
		MCDB	1.5	1.7	6.6	13.4	20.6	23.8	25.2	24.6	22.4	16.5	9.4	3.4
	10%	WB	-0.9	-0.9	2.3	6.6	13.6	18.3	20.1	19.7	18.0	12.1	6.0	1.2
		MCDB	-0.1	0.3	4.2	10.9	18.4	22.0	24.0	22.9	20.7	14.5	7.5	1.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.0	10.7	10.9	13.1	11.6	11.0	10.1	9.8	7.6	6.4	7.1
		MCDBR	8.3	9.6	13.0	17.4	18.6	15.1	13.5	12.7	12.5	11.9	9.5	6.9
		MCWBR	7.8	7.9	8.6	9.7	9.1	7.0	5.6	5.6	6.5	7.2	7.4	6.4
	5% WB	MCDBR	8.1	8.7	11.8	15.5	15.9	12.6	11.6	10.5	10.8	10.8	8.9	6.8
		MCWBR	7.8	7.6	8.2	9.7	9.0	6.8	5.8	5.4	6.5	7.3	7.4	6.5
Clear-Sky Solar Irradiance	taub	0.270	0.286	0.322	0.358	0.369	0.400	0.402	0.394	0.372	0.338	0.304	0.272	
	taud	2.317	2.252	2.268	2.340	2.391	2.333	2.351	2.369	2.409	2.492	2.501	2.383	
	Ebn at Noon	826	884	899	897	896	866	859	851	839	821	788	784	
	Edh at Noon	74	98	113	117	115	123	119	112	99	79	64	63	
All-Sky Solar Radiation	RadAvg	1.27	2.22	3.68	4.55	5.21	5.64	5.60	4.78	3.63	2.17	1.18	0.94	
	RadStd	0.18	0.29	0.38	0.37	0.46	0.41	0.43	0.31	0.39	0.27	0.14	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page